

Job Market Signaling

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Job Market Model

- Many identical firms that can hire a worker.
- Worker's marginal productivity: $\theta \sim F(\cdot)$.
- Worker's outside option $r(\theta)$.
- Unobservable θ by the firms.

Bayesian Nash Equilibrium

Let $\Theta(w^*)$ denote the types that participate in the labor market in equilibrium. Then BNE is defined by $(w^*, \Theta(w^*))$ where

$$\Theta^* = \{\theta : r(\theta) \leq w^*\}$$

$$w^* = E[\theta | \theta \in \Theta^*]$$

Example: Market Unraveling

- Let $r(\theta) = \alpha\theta$.
- $\theta \sim U[0, 2]$.
- Then,

$$E[\theta|r(\theta) \leq w] = E[\theta|\theta < w/\alpha] = \frac{w}{2\alpha}$$

- If $\alpha \leq \frac{1}{2}$, then $E[\theta|r(\theta) \leq w] \geq w$. Therefore,

$$w^* = E[\theta] = 1 > r(\theta) \quad \forall \theta$$

- If $\alpha > \frac{1}{2}$, then $E[\theta|r(\theta) \leq w] < w$.

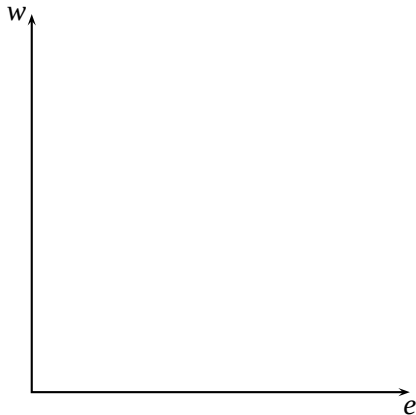
Job Market Signaling: Spence (1973)

- It is conceivable that high ability workers may take actions to differentiate themselves from low ability workers.
- Let's consider two types: $\{\theta_H, \theta_L\}$.
- $F(\cdot)$ is binary with $Prob(\theta = \theta_H) = \lambda$.
- Before entering the labor market, a worker may get some education that is observable.
- Cost of education $c(e, \theta)$:

$$c(0, \theta) = 0, c_e(e, \theta) > 0, c_{ee}(e, \theta) < 0, c_\theta(e, \theta) < 0, c_{e\theta}(e, \theta) < 0$$

- Simplifying assumptions:
 - education has no impact on productivity: $u(w, e|\theta) = w - c(e, \theta)$.
 - $r(\theta) = 0$ for all θ .

Single Crossing Property



Timing

- 1 Nature moves and determines worker's type;
 - 2 The worker observes his type and chooses his education level;
 - 3 The firms observe the worker's education level and compete for the worker by making wage offers.
- the last stage of the game is assumed to be perfectly competitive wage setting game so that the wage is equal to the expected productivity of the worker (ex: Bertrand competition).
 - We want to characterize PBE.

Perfect Bayesian Equilibrium

- For PBE, the following requirements need to be satisfied:
 - 1 The worker's education choice is optimal given the firms' equilibrium wage offer and his type.
 - 2 The firms' wage offers constitute BNE of the simultaneous move wage offer game given their belief $\mu(e) = \text{Prob}(\theta_H|e)$.
 - 3 $\mu(e)$ is derived from the worker's strategy using Bayes' rule where possible.
- The firm's wage offers are:

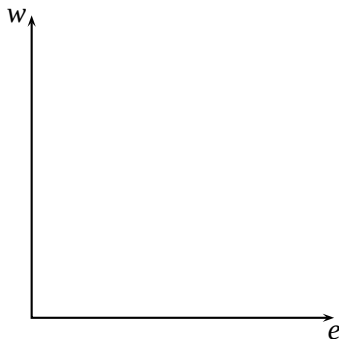
$$w^*(e) = \mu(e)\theta_H + (1 - \mu(e))\theta_L$$

Pooling Equilibria

- In a pooling equilibrium, the high and the low type choose the same education level (e_p):

$$\mu(e_p) = \lambda$$

$$w^*(e_p) = \lambda\theta_H + (1 - \lambda)\theta_L = E[\theta]$$



Pooling Equilibria

- The highest equilibrium level of education sustainable under pooling equilibrium $\bar{e}_p$ is characterized by:

$$E[\theta] - c(\bar{e}_p, \theta_L) = \theta_L$$

- Any education level $\hat{e} \leq \bar{e}_p$ can be sustained as PBE with the following belief stricture:

$$\mu(e) = \begin{cases} 0 & \text{if } e < \hat{e} \\ \lambda & \text{if } e \geq \hat{e} \end{cases}$$

$$w^*(e) = \begin{cases} \theta_L & \text{if } e < \hat{e} \\ E[\theta] & \text{if } e \geq \hat{e} \end{cases}$$

Example

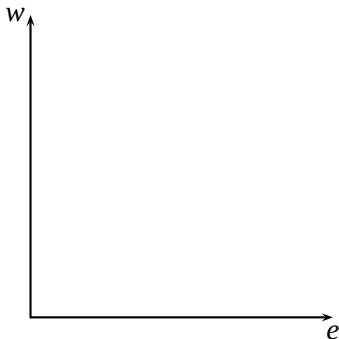
- Let $c(e, \theta) = \frac{e^2}{\theta}$, $\theta_L = 1$, $\theta_H = 2$, and $\lambda = \frac{1}{2}$.

Separating Equilibria

- The low type must find it too costly to mimic the high type.
- It must be true that

$$w^*(e_s(\theta)) = \theta$$

$$e_s(\theta_L) = 0$$



Separating Equilibria

- No worker should have incentives to deviate from e^* .
- For the low type, this implies that $e^* \geq \underline{e}_s$ where:

$$\theta_H - c(\underline{e}_s, \theta_L) = \theta_L$$

- For the high type, this implies that $e^* \leq \bar{e}_s$

$$\theta_H - c(\bar{e}_s, \theta_H) = \theta_L$$

- Let $c(e, \theta) = \frac{e^2}{\theta}$, $\theta_L = 1$, $\theta_H = 2$, and $\lambda = \frac{1}{2}$.

Separating Equilibria

- Thus, any $\hat{e} \in [\underline{e}_s, \bar{e}_s]$ can be supported as a separating equilibrium by defining:

$$\mu(e) = \begin{cases} 0 & \text{if } e \neq \hat{e} \\ 1 & \text{if } e = \hat{e} \end{cases}$$

$$w^*(e) = \begin{cases} \theta_L & \text{if } e \neq \hat{e} \\ \theta_H & \text{if } e = \hat{e} \end{cases}$$

Equilibrium Refinement

- There is multiplicity of pooling and separating equilibria (some more costly than others).
- Are some equilibria more reasonable than others?
- PBE allows us to assign any beliefs to 0 probability events.
- This includes $Prob(\theta_L | e > \underline{e}_s) = 1$.

Cho-Kreps Intuitive Criterion

Equilibrium-dominated education level

An education level e_i is equilibrium-dominated for θ_i if θ_i 's equilibrium payoff given $(e_i^*, w^*(e_i^*), \mu^*(e_i^*))$ specified by the PBE exceeds the payoff under e_i for any possible $\mu(e_i)$:

$$u^*(e_i^*, \mu^*(e_i^*)|\theta_i) > \max_{\mu(e_i)} u_i(e_i, \mu(e_i)|\theta_i)$$

Cho-Kreps Intuitive Criterion

If e_i is equilibrium dominated, then the firms' belief should be $Prob(\theta_i|e_i) = 0$ as long as e_i is not equilibrium dominated by all types.

- The Intuitive Criterion requires that $Prob(\theta_L|e) = 0$ for $e \in (\underline{e}_S, \bar{e}_S]$.
- Thus, the only separating equilibrium that survives involves $\mu(e) = 1$ for all $e \geq \underline{e}_S$.

No Pooling Equilibrium under the Intuitive Criterion

- Let e^* be the education level under a pooling equilibrium.
- Define e' as:

$$E[\theta] - c(e^*, \theta_L) = \theta_H - c(e', \theta_L)$$

- For any $e > e'$, $\mu(e) = 1$.
- No deviation incentives for θ_H requires:

$$E[\theta] - c(e^*, \theta_H) \geq \theta_H - c(e', \theta_H)$$

No Pooling Equilibrium under the Intuitive Criterion

- Rearranging terms:

$$c(e', \theta_H) - c(e^*, \theta_H) \geq c(e', \theta_L) - c(e^*, \theta_L)$$

$$\int_{e^*}^{e'} c_e(e, \theta_H) de \geq \int_{e^*}^{e'} c_e(e, \theta_L) de$$

- However, this inequality contradicts $c_{e\theta}(e, \theta) < 0$ implying that no pooling equilibrium exists.

Positive outside option

- Let $r(\theta_H) = r(\theta_L) = r > 0$
- Suppose that $\theta_L < E[\theta] < r < \theta_H$.
- With no ability to signal, both types stay at home.
- Under a separating equilibrium:
 - low type stays at home.
 - high type acquires education since for $\underline{e}_s$

$$r = \theta_H - c(\underline{e}_s, \theta_L) < \theta_H - c(\underline{e}_s, \theta_H)$$

- Thus, the possibility of signaling can prevent unraveling and assign the workers to the task at which their productivity is maximized.
- References: Mas-Colell- Ch 13, Gibbons Ch. 4.